



JSS
SCIENCE AND
TECHNOLOGY
UNIVERSITY
MYSURU

- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
- Governed by the Grant-in-Aid Rules of Government of Karnataka
- Identified as lead Institution for World Bank Assistance under TEQIP Scheme



DEPARTMENT OF CIVIL ENGINEERING

SCHEME & SYLLABUS

First Semester

(From the Academic Year 2022-23)

Course Title	Engineering Mechanics			Course Type		3:0:0			
Course Code	22CV110/ 22CV210	Credits		Class		1 st Year B.E.			
Course Structure	TLP	Credits	Contact Hours	Total Number of Classes / Semester		Assessment in Weightage and Marks			
	Theory	3	3	Theory	Practical	CIE		SEE	
	Practical	0	0	39	0	Weightage	40%	Weightage	60%
	Tutorial	0	0	0	0	Max. Marks	40 Marks	Max. Marks	60 Marks
	Total	3	3	39	0	Min. Marks	20 Marks	Min. Marks	25 Marks

Course Overview: Engineering Mechanics is a fundamental course which deals with analyzing the effect of force system on rigid bodies, location of centroid and moment of inertia of plane lamina. The course enhances analyzing and problem solving skills in students of all branches of engineering.

Pre requisites: —

Course Objective:

1. To analyse problems in engineering subjected to a force system and to compute the net effect.
2. To analyse the equilibrium of rigid bodies and structures.
3. To determine the geometric properties of plane sections.

Course Outcome:

CO	Course Outcome	Highest Level of Cognitive Domain
CO1	Explain fundamental concepts, principles, laws, idealizations, theorems of Engineering Mechanics.	L1, L2
CO2	Compute resultant for given concurrent and non-concurrent force system.	L3
CO3	Solve the force system for equilibrium condition.	L3
CO4	Determine support reactions for statically determinate systems.	L3
CO5	Locate the centroid of plane figures and to compute the second moment of areas of standard sections.	L3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Articulation:

POs → COs ↓	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	-	-	-	-	-	-	-
CO3	3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-
CO5	3	3	-	-	-	-	-	-	-	-	-	-

High – 3, Medium – 2, Low – 1

Course Content / Syllabus:

UNIT No.	Content	Lecture Hours
UNIT 1	Introduction to Engineering Mechanics: Basic idealizations: Particle, Continuum, Rigid body and Point force; Newton's laws of motion, Definition of Force, Introduction to SI units, Elements of a force, Classification of force and force systems; Principle of physical independence of forces, Principle of superposition of forces, Principle of transmissibility of forces; Moment of a force, couple, moment of a couple, characteristics of a couple, Equivalent force: couple system; Resolution of a force, Composition of forces; Numerical problems on resolution of forces, moment of forces and couples and equivalent force couple system.	08
UNIT 2	Composition of forces: Definition of Resultant; Composition of coplanar - concurrent force system, Principle of resolved parts; Numerical problems on composition of coplanar-concurrent force systems. Composition of Coplanar: non-concurrent force system, Varignon's principle of moments; Numerical problems on composition of coplanar non-concurrent force systems.	09
UNIT 3	Equilibrium of forces: Definition of Equilibrant; Conditions of static equilibrium for different force systems, Lami's theorem; Numerical problems on equilibrium of coplanar – concurrent force system. Support Reactions: Types of Beams, Types of supports, statically determinate beams, Numerical problems on equilibrium of coplanar – non-concurrent force system and support reactions for statically determinate beams.	10
UNIT 4	Centroid: Location of centroid of Plane Figures: Rectangle, Triangle, Semicircle, Quadrant of a circle. Numerical problems on simple built up Sections.	06
UNIT 5	Moment of Inertia: Determination of Moment of inertia of plane figures: Rectangle, Triangle, Circle and Semi circle. Polar moment of inertia, Radius of gyration, Perpendicular axis theorem and Parallel axis theorem. Numerical problems on simple built up Sections.	06

Text Books:

- Beer, F.P. and Johnston Jr., E.R. (1982), Mechanics for Engineers, Vol. 1 – Statics, *McGraw-Hill Book Company, New York*.
- Rahman S.S. and Madhava Rao, V. (2006), Elements of Civil Engineering and Engineering Mechanics, *Sanguine Technical Publishers, Bengaluru*.

Reference Books:

1. Merium, J.L. and Kraige, L.G. (2006), Engineering Mechanics, Vol. 1 – Statics, 3rd Edition, John Wiley and Sons Inc., New York.
2. Kumar, K.L. (2008), Engineering Mechanics, 3rd Revised Edition, Tata McGraw-Hill Publishing Company, New Delhi.
3. Boresi, A.P. and Schmidt, R.J. (2000), Engineering Mechanics, CL-Engineering, USA.

Web/Digital resources:

1. <https://nptel.ac.in/courses/112/106/112106286/>
2. VTU – e-Learning

Evaluation Scheme:**Continuous Internal Evaluation – CIE**

Event	Event Type	Marks Allotted	Duration
CIE – 1	Written Test – 1	30	1 Hour
CIE – 2	Event	20	1 Hour
CIE – 3	Written Test – 2	30	1 Hour

Note:

1. The written test 1 & 2 (CIE – 1 & 3) both will be conducted for 30 marks each in 1 hour duration and the marks scored will be reduced proportionately to 15 marks each.
2. The Event (CIE – 2) will be conducted for 20 marks and the marks scored will be reduced proportionately to 10 marks.
3. The Event (CIE – 2) will be skill based assessment such as Seminars / Technical talks / Case study / hands-on activity / Mini projects / Sci-tech activity / Data analysis.
4. A student must score on an average of 50% i.e., 20 marks out of 40 from all the events (CIE - 1, 2, 3) to gain the eligibility to appear for SEE.

Semester End Examination – SEE

Event	Event Type	Marks Allotted	Duration
SEE	Written Examination	100	3 Hours

Note:

1. The SEE will be conducted for 100 marks and the marks scored will be proportionately reduced to 60 marks. A student must score a minimum of 25 marks out of 60 in the SEE to pass.
2. SEE Question paper will be set for 100 marks and will have two parts (Part-A & Part-B). Questions under Part-A are compulsory and questions under Part-B will have internal choices.